OLYMPIC SPORTS PERFORMANCE ANALYSIS

DOCUMENT 02

Data Analysis and Methodology

Data structure, analytical questions, measures, and validation

Academic Project Documentation

Purpose and Scope

This document records the project's approach based on the project context available. Compare all descriptions and counts with the final Tableau workbook and dataset before academic submission.

Data note: Important: project context describes the workbook as using demonstration/synthetic data. Treat findings as findings from this dataset, not verified historical Olympic facts, unless independently verified.

1. Purpose of the Analysis

The analysis organizes records into comparable units so medal outcomes, participation, country-year performance, sport categories, and regional summaries can be explored consistently.

2. Workbook Data Structure

- athletes — athlete-level records for participation views.
- medal_results — records used to summarize medal outcomes.
- country_year_summary — country-level summaries by year.
- investment — values used in the illustrative investment-versus-medals view.
- annual_medal_trends — time-series summaries.
- host_medal_reference — reference information for host-country comparisons.
- regional_efficiency — regional comparison metrics.
- sport_category_metrics — summaries by sport or sport category.
- reference_metrics — reference values used by selected views.
- data_dictionary — field descriptions and interpretation notes.

3. Reported Dataset Scale

Project context describes approximately 4,652 athlete records, 3,164 medal-result records, and 1,010 country-year summaries, covering 1992–2026, 60 countries, and 50 sports. These approximate figures should be verified against the actual workbook before submission.

4. Analysis Questions

- How are medal results distributed across countries?
- How do medal totals change over the years in the available data?
- How does athlete participation compare with medal outcomes?
- What differences appear across regions and sport categories?
- How do host-country results compare with a selected reference?
- What relationship is visible between investment and medal outcomes in the included data?

5. Analytical Measures

The dashboard includes or proposes medal totals, athlete counts, average medals per athlete, country metrics, regional efficiency, and investment-versus-medal comparisons. The exact formula for each calculated field should be documented. Average medals per athlete, for example, requires a clear denominator and a decision about how athletes with no medals are treated.

6. Suggested Workflow

- Inspect source tables and the data dictionary.

- Check missing values, duplicates, field types, and category spelling.
- Confirm each table's level of detail before joining or relating tables.
- Validate aggregation logic to avoid double-counting athletes or medals.
- Compare summary metrics with hand-calculated samples.
- Build charts after measures have been checked.

7. Data Quality and Interpretation

Document whether data is official, compiled, simulated, or synthetic. State sources and licenses where applicable. Use consistent labels for countries, regions, sports, and years. Correlation is not proof of causation, and trends in a demonstration dataset should not be presented as verified historical facts.